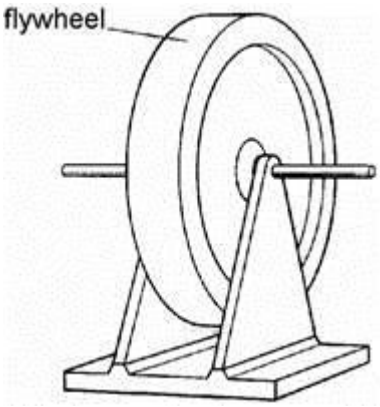
Q1.

A flywheel is a rotating wheel which is used to store energy.



(a) Energy must be transferred to a flywheel to make it rotate. How is the energy in the rotating flywheel classified?
Tick the correct box.

as chemical energy ☐

as kinetic energy ☐

as potential energy ☐

as thermal energy ☐

1 mark

(b) A flywheel is rotating at a high speed. No energy is being supplied to it. The flywheel is used to turn a dynamo, and the energy from the dynamo is used to light a bulb.

(i) The bulb is left connected until the flywheel stops rotating. Not all the energy stored in the flywheel is transferred to the bulb. Some of it is lost. Give **two** places from which it is lost, and explain how it is lost.

1. .....

.....

2. .....

.....

2 marks

(ii) The experiment is repeated using a different bulb which gives out more energy each second. Compared to the first light bulb, describe how the second light bulb will affect the motion of the flywheel, and explain your answer.

.....

.....

.....

.....

2 marks
Maximum 5 marks

Q2.

A pupil has two well-insulated containers of water at 20° C. She takes two identical blocks of aluminium, both at 100° C, and puts one into container A and the other into container B. The temperature of the water rises in each container until it reaches a steady value. Her results are shown in the table.

container	temperature of water in °C at the start	temperature of block in °C at the start	volume of water in cm^3	final temperature of water in °C
A	20	100	500	44
B	20	100	1000	34

(a) Explain why the final temperature of the water is lower in container B.

.....

.....

1 mark

(b) What is the final temperature of each aluminium block?

The block in container A is at .....°C.

The block in container B is at .....°C.

1 mark

(c) Which of the identical aluminium blocks transferred more energy to the water? Explain your reasoning.

.....

.....

1 mark
Maximum 3 marks

Q1.

(a) as kinetic energy ✓

if more than one box is ticked, award no mark

1 (L6)

(b) (i) **both the place and the method by which energy is lost are required for each mark**

answers may be in either order

- from the axle **or** bearings by heat **or** sound

accept 'from the bearing by friction'
or *'the bearings get hot*
or *'from the axle when it squeaks'*

1 (L7)

- from the wires by heat

accept 'the wires get hot'

*accept 'from the dynamo as heat **or** sound'*
or *'from the dynamo when it gets hot **or** squeaks'*
*do **not** accept 'goes into the air as heat **or** sound'*

1 (L7)

(ii) it slows down more quickly

accept 'it rotates for a shorter time'

*do **not** accept 'it slows down'*

1 (L7)

because it transfers energy to the bulb more quickly

accept 'because it transfers more energy to the bulb'
*do **not** accept 'because it transfers energy to the bulb'*

1 (L7)

[5]

Q2.

(a) the energy had to be shared by more particles **or** water

accept 'there was more water to heat up'
or *'the heat is spread out more'*

*do **not** accept 'there was more water'*

1 (L7)

(b) A 44

B 34

both answers are required for the mark

1 (L7)

(c) **for the mark to be awarded, the correct block must be identified and the correct reason given**

any **one** from

- the block in B, because its temperature dropped by the larger amount

  both answers are required for the mark
  *do **not** accept 'B, because there was more water to heat up'*

- the block in B, because this block ended up at a lower temperature

1 (L7)

[3]